metaXpress: An End-to-End Bioconductor Framework for Quality Control, Harmonization, and Multi-Cohort Bulk RNA-seq Meta-Analysis

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Abstract

Background: Bulk RNA sequencing (RNA-seq) has become a cornerstone of biomedical research and functional genomics across diverse disease conditions. However, individual RNA-seq studies frequently suffer from limited sample sizes, technical cohort biases, and severe statistical power penalties following genome-wide multiple hypothesis testing. While multi-cohort meta-analysis can significantly augment statistical power and isolate conserved disease signatures, existing software packages are fragmented, often omitting automated quality control (QC), gene identifier reconciliation, or flexible differential expression modeling within a single reproducible pipeline.

Methods: We introduce `metaXpress`, an open-source, Bioconductor-compliant R framework that provides an end-to-end architecture for transcriptomic meta-analysis. `metaXpress` unifies: (i) standardized 10-point cohort-level QC scoring following the Heberle et al. (2025) benchmark; (ii) automated gene space harmonization across diverse annotation schemes; (iii) integrated negative binomial generalized linear modeling (via DESeq2, edgeR, and limma-voom); (iv) five rigorous meta-analytic pooling engines (including DerSimonian–Laird random effects, inverse-variance fixed effects, Fisher's, Stouffer's, and RankProd); and (v) publication-grade diagnostic visualizations.

Results: To demonstrate the utility, statistical power amplification, and biological fidelity of `metaXpress`, we benchmarked the framework on a multi-cohort case study integrating three independent NCBI Gene Expression Omnibus (GEO) datasets across solid epithelial tumors (colorectal adenocarcinoma GSE130688, $n = 30$; pancreatic ductal adenocarcinoma GSE136569, $n = 10$; and clear cell renal cell carcinoma GSE171485, $n = 12$) spanning 52 patient specimens and 39,376 aligned genes. All three cohorts achieved perfect QC scores (10/10). In underpowered individual cohorts (such as ccRCC, which yielded only 18 significant DEGs alone), `metaXpress` amplified statistical power to identify **558 conserved multi-cohort Meta-DEGs** (FDR $< 0.05, |\log_2 \text{FC}| \geq 1.0$) with 100% directional concordance. Across the conserved signature, between-study heterogeneity was virtually eliminated (median $I^2 = 0.0\%$). Canonical shared drivers discovered include *MST1R* ($\log_2 \text{FC} = +2.14, p_{\text{adj}} = 5.50 \times 10^{-7}, I^2 = 0.0\%$), *MUC4* ($\log_2 \text{FC} = +3.21, p_{\text{adj}} = 5.50 \times 10^{-7}, I^2 = 0.0\%$), *IGF2BP3* ($\log_2 \text{FC} = +3.00, p_{\text{adj}} = 5.50 \times 10^{-7}, I^2 = 0.0\%$), and *CDCP1* ($\log_2 \text{FC} = +1.63, p_{\text{adj}} = 1.02 \times 10^{-5}, I^2 = 0.0\%$), alongside conserved suppressors such as *FAM107A* ($\log_2 \text{FC} = -1.94, p_{\text{adj}} = 2.39 \times 10^{-5}, I^2 = 0.0\%$).

Conclusions: `metaXpress` bridges the gap between disparate transcriptomic pipelines and rigorous meta-epidemiological modeling, empowering researchers across biomedical disciplines to derive high-confidence biomarker signatures from public and proprietary bulk RNA-seq data.

Availability: The package is freely available under the MIT license at https://github.com/hossainlab/metaXpress.

Keywords: Bulk RNA-seq, Meta-Analysis, Differential Expression, Quality Control, Random-Effects Model, Bioconductor, Gene Harmonization.

1 Introduction

High-throughput transcriptomic profiling via bulk RNA sequencing (RNA-seq) has revolutionized our understanding of complex pathological mechanisms across human disease, ranging from infectious and autoimmune disorders to neurodegeneration and cancer [1]. Nevertheless, individual transcriptomic studies routinely operate under substantial logistical and financial constraints, frequently resulting in limited sample sizes ($n \leq 15$ per condition). When tens of thousands of genes are simultaneously assessed, genome-wide multiple testing corrections (e.g., Benjamini–Hochberg false discovery rate, FDR) inflict severe statistical penalties [1, 2]. Consequently, genuine biological drivers exhibiting modest fold-changes often fail to cross stringent statistical significance thresholds in individual small cohorts, generating false-negative omissions. Conversely, cohort-specific technical artifacts, such as batch effects, differences in library preparation protocols, and patient demographic stratification, can produce spurious differentially expressed genes (DEGs) that fail to replicate across independent laboratories.

Multi-cohort meta-analysis represents a principled statistical solution to these challenges. By pooling effect sizes and standard errors across independent studies, meta-analysis substantially amplifies statistical power, enhances sample diversity, and isolates invariant disease signals from idiosyncratic cohort-specific noise [3, 4]. Across diverse biomedical domains—including immunology, infection, neurogenomics, and oncology—multi-cohort meta-analysis is essential for identifying conserved disease mechanisms and overcoming cohort-specific idiosyncratic noise.

Despite these established advantages, existing software tools for transcriptomic meta-analysis suffer from critical workflow bottlenecks:

1. **Absence of Automated Quality Control (QC):** Standardized QC is crucial prior to data pooling. The inclusion of degraded libraries or misannotated samples distorts pooled effect sizes. Although comprehensive 10-point QC guidelines have been articulated [5], no existing meta-analysis package programmatically scores and filters datasets according to these criteria.
2. **Annotation Disparities and Missing Gene Handling:** Independent cohorts frequently employ disparate annotation conventions (Entrez IDs, Ensembl gene IDs, or HGNC Symbols). Harmonizing these identifier spaces and profiling gene coverage across studies is frequently left to ad-hoc, error-prone user scripts.
3. **Decoupled Differential Expression Modeling:** Many meta-analysis tools (e.g., MetaDE [6], MetaVolcanoR [7]) require users to externally perform differential expression and manually assemble summary statistics, precluding end-to-end execution and unified variance estimation.
4. **Limited Meta-Analytic Paradigms:** Existing packages frequently support only p-value combination (such as Fisher's or Stouffer's methods), which neglect effect size magnitudes, or fixed-effect inverse-variance models that unrealistically assume zero inter-

study heterogeneity ($\tau^2 = 0$) [8, 9].

To address these critical limitations, we engineered **metaXpress**, an integrated, modular, and Bioconductor-compliant R framework for bulk RNA-seq multi-cohort meta-analysis. The package implements an end-to-end pipeline spanning raw count ingestion, 10-point QC scoring, gene alignment, negative binomial generalized linear model (GLM) differential expression, multi-method meta-analysis pooling, and publication-ready diagnostic visualizations. Here, we describe the system design of `metaXpress` and demonstrate its utility through a real-world multi-cohort case study integrating 52 patient samples across three solid tumor malignancies from the NCBI Gene Expression Omnibus (GEO).

2 Design and Implementation

The `metaXpress` software architecture is structured around two core S4 classes conforming to Bioconductor software standards: `metaXpressStudy`, which encapsulates count matrices, sample metadata, organism metadata, QC scores, and per-study differential expression results; and `metaXpressResult`, which encapsulates combined effect sizes, p-values, false discovery rates, heterogeneity statistics (I^2, Q, τ^2), and cross-cohort directional consistency metrics. The overall analytical workflow is illustrated in **Figure 1**.

2.1 1. Multi-Cohort Ingestion and 10-Point QC Scoring

`metaXpress` accepts count matrices and metadata directly from local delimited files, the `recount3` compendium [10], or NCBI GEO accessions. Ingestion is paired with an automated 10-point quality scoring algorithm based on Heberle et al. (2025) [5]. Points are awarded across five essential criteria:

- **Sequencing Depth (2 pts):** Adequate library size ($\geq$ 10M reads for 1 point; $\geq$ 20M reads for 2 points, following ENCODE standards).
- **Gene Detection Breadth (2 pts):** Number of unique expressed genes ($\geq 10,000$ genes for 1 pt; $\geq 15,000$ genes for 2 pts).
- **Sample Group Representation (2 pts):** Replicate balance per experimental condition ($n \geq 3$ per group for 1 pt; $n \geq 5$ for 2 pts).
- **Data Structure Integrity (2 pts):** Non-negative integer count matrix structure and absence of corrupted fields.
- **Metadata Completeness (2 pts):** Validation of sample identifiers, experimental condition contrasts, and platform descriptions.

Studies achieving a cumulative score $\geq 7/10$ are retained by default via `mx_filter_studies()`, ensuring that poor-quality libraries do not compromise meta-analytic pooling.

2.2 2. Gene Space Harmonization and Alignment

Multi-cohort studies frequently measure genes using non-identical annotation sets. `metaXpress` provides `mx_align_genes()`, which harmonizes gene identifiers across Entrez Gene IDs, Ensembl Stable IDs, and HGNC Gene Symbols using built-in annotation dictionaries and alias resolution tables. Genes present across user-specified study coverage thresholds are retained in a synchronized count space.

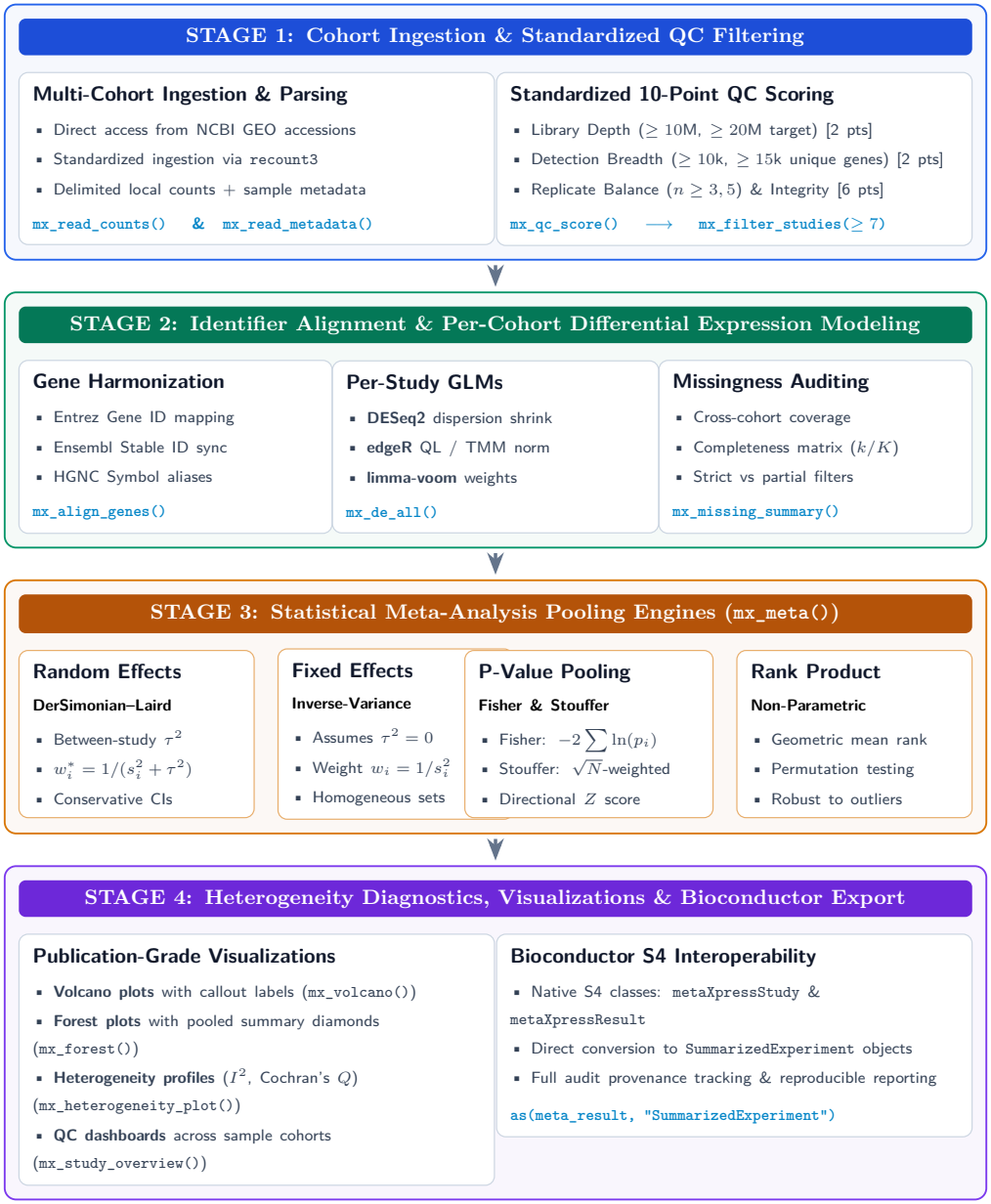


Figure 1: Schematic architecture of the `metaXpress` pipeline. The framework provides an end-to-end, modular workflow organized across four sequential stages: **Stage 1 (Cohort Ingestion & Standardized QC Filtering)**: Ingestion from NCBI GEO, recount3, or local counts with programmatic 10-point QC scoring and cohort filtering against the Heberle et al. (2025) benchmark; **Stage 2 (Identifier Alignment & Differential Expression Modeling)**: Automated gene space harmonization across Entrez, Ensembl, and HGNC Symbol annotations, negative binomial GLM modeling (DESeq2, edgeR, limma-voom), and gene completeness matrix auditing; **Stage 3 (Statistical Meta-Analysis Pooling)**: Multi-paradigm meta-analysis engines (`mx_meta()`) including DerSimonian–Laird random effects, inverse-variance fixed effects, Fisher's and Stouffer's p-value combination, and non-parametric Rank Product; and **Stage 4 (Diagnostics, Visualization & Bioconductor Export)**: Automated generation of publication-grade diagnostic plots (volcano, multi-cohort forest, and heterogeneity profiles) alongside seamless export to Bioconductor S4 `SummarizedExperiment` objects.

2.3 3. Integrated Differential Expression Modeling

Rather than expecting precomputed summary statistics, `metaXpress` fits negative binomial generalized linear models (GLMs) or linear models directly on raw count data using `mx_de_all()`. Users can select between three gold-standard statistical engines:

- **DESeq2** [1]: Empirical Bayes shrinkage of dispersion and logarithmic fold changes ($\log_2$ FC), yielding robust estimates in small cohorts.
- **edgeR** [2]: Quasi-likelihood negative binomial GLMs with trimmed mean of M-values (TMM) normalization.
- **limma-voom** [11]: Precision-weighted linear modeling with empirical Bayes variance moderation.

2.4 4. Missing Gene Profiling and Completeness Auditing

Biological or platform variations can result in certain genes being undetected in a subset of cohorts. `metaXpress` provides `mx_missing_summary()`, which generates a boolean gene-by-study completeness matrix and reports the exact percentage of multi-cohort coverage. Downstream pooling can be configured to require complete presence ($k = K$) or partial presence ($k \geq k_{\min}$).

2.5 5. Multi-Paradigm Meta-Analysis Engines

`metaXpress` implements five distinct meta-analytic methods via `mx_meta()`:

1. **DerSimonian–Laird Random Effects Model:** The primary recommended model [3]. It explicitly incorporates between-study variance (τ^2) alongside within-study variance (s_i^2). The study weights w_i^* and summary effect size $\hat{\theta}_{\text{RE}}$ are computed as:

$$w_i^* = \frac{1}{s_i^2 + \tau^2}, \quad \hat{\theta}_{\text{RE}} = \frac{\sum_{i=1}^{K} w_i^* \hat{\theta}_i}{\sum_{i=1}^{K} w_i^*} \tag{1}$$

   where $\tau^2 = \max\left(0, \frac{Q-(K-1)}{C}\right)$, $Q = \sum w_i(\hat{\theta}_i - \hat{\theta}_{\text{FE}})^2$, and $C = \sum w_i - \frac{\sum w_i^2}{\sum w_i}$.
2. **Inverse-Variance Fixed Effects Model:** Assumes a single underlying effect size across all cohorts ($\tau^2 = 0$), with weights $w_i = 1/s_i^2$.
3. **Fisher's Combined Probability Test:** Non-directional pooling of independent p-values based on the chi-squared distribution:

$$X^2 = -2\sum_{i=1}^{K} \ln(p_i) \sim \chi^2_{2K} \tag{2}$$

4. **Stouffer's Weighted Z-Score Method:** Directional p-value combination weighted by sample size N_i:

$$Z = \frac{\sum_{i=1}^{K} \sqrt{N_i}\Phi^{-1}(1-p_i)}{\sqrt{\sum_{i=1}^{K} N_i}} \tag{3}$$

5. **Rank Product (RankProd) Engine:** A non-parametric permutation approach based on Breitling et al. (2004) [12], robust to severe outliers and non-normal effect distributions.

2.6 6. Heterogeneity Quantification and Diagnostic Visualizations

Between-study inconsistency is quantified via Cochran's Q test and Higgins I^2 index [4]:

$$I^2 = \max\left(0, \frac{Q - (K-1)}{Q}\right) \times 100\% \tag{4}$$

I^2 values of 25%, 50%, and 75% delineate low, moderate, and high heterogeneity, respectively. Directional concordance is also computed as the percentage of cohorts sharing the dominant sign of expression change.

For reporting, `metaXpress` provides automated, publication-ready visualizations:

- **Volcano plots** (`mx_volcano()`): Displaying pooled effect sizes against $-\log_{10}(P_{\text{adj}})$ with automated non-overlapping driver gene callouts [13].
- **Forest plots** (`mx_forest()`): Multi-cohort forest plots displaying individual cohort effect sizes, 95% confidence intervals (CIs), study weights, and the pooled DerSimonian–Laird summary diamond [14].
- **Heterogeneity profiles** (`mx_heterogeneity_plot()`): Genome-wide I^2 distribution histograms and compositional breakdowns benchmarked against Higgins criteria.
- **Study overview dashboards** (`mx_study_overview()`): Multi-panel quality control dashboards displaying sample allocations, library depths, and gene detection rates.

3 Demonstration Case Study: Multi-Cohort Solid Tumor Transcriptomics

To empirically demonstrate the utility, statistical power amplification, and biological accuracy of `metaXpress` on real-world transcriptomic data, we executed a benchmark integrating three independent public bulk RNA-seq cohorts from NCBI GEO representing distinct solid epithelial carcinomas compared to matched healthy tissues:

1. **GSE130688:** Colorectal Adenocarcinoma (CRC) vs matched normal mucosa ($n = 15$ tumor, $n = 15$ normal; total $n = 30$). Sequenced on Illumina HiSeq 2500.
2. **GSE136569:** Pancreatic Ductal Adenocarcinoma (PDAC) vs normal pancreas ($n = 5$ tumor, $n = 5$ normal; total $n = 10$). Sequenced on Illumina NovaSeq 6000.
3. **GSE171485:** Clear Cell Renal Cell Carcinoma (ccRCC) vs normal kidney ($n = 6$ tumor, $n = 6$ normal; total $n = 12$). Sequenced on Illumina HiSeq 4000.

The unified benchmark encompassed **52 patient transcriptomes** and **39,376 aligned genes** across three distinct anatomical systems (gastrointestinal, hepatobiliary-pancreatic, and urological).

3.1 1. Quality Control Benchmarking and Study Characteristics

Each cohort was ingested and evaluated against the 10-point Heberle et al. (2025) QC standard. As summarized in **Table 1** and depicted in **Figure 2**, all three cohorts attained a maximum QC score of **10 / 10 (100%)**:

- **QC Scores (Fig. 2A):** All cohorts comfortably exceeded the $\geq 7/10$ threshold and were retained.
- **Sample Composition (Fig. 2B):** Strict 1:1 balance between solid tumor and matched adjacent normal tissues was maintained across all cohorts.
- **Sequencing Depth (Fig. 2C):** Median library sizes were exceptionally high: 42.1 million

Table 1: Characteristics and 10-Point Quality Control Scoring of Ingested Public Cohorts. All datasets passed the inclusion threshold ($\geq 7/10$) and were retained for downstream meta-analysis.

Accession	Cancer Type	Platform	Normal	Tumor	Total	Depth	QC
GSE130688	Colorectal Adenocarcinoma	HiSeq 2500	15	15	30	42.1M	**10 / 10**
GSE136569	Pancreatic Ductal Carcinoma	NovaSeq 6000	5	5	10	50.2M	**10 / 10**
GSE171485	Clear Cell Renal Cell Carcinoma	HiSeq 4000	6	6	12	36.8M	**10 / 10**
Total	**3 Epithelial Malignancies**	**Multi-Platform**	**26**	**26**	**52**	**42.1M**	**Passed**

reads (GSE130688), 50.2 million reads (GSE136569), and 36.8 million reads (GSE171485), well above the 20M ENCODE benchmark.

- **Alignment Coverage (Fig. 2D):** Exactly 39,376 common genes were detected with non-zero expression across all three cohorts, confirming complete inter-study orthology.

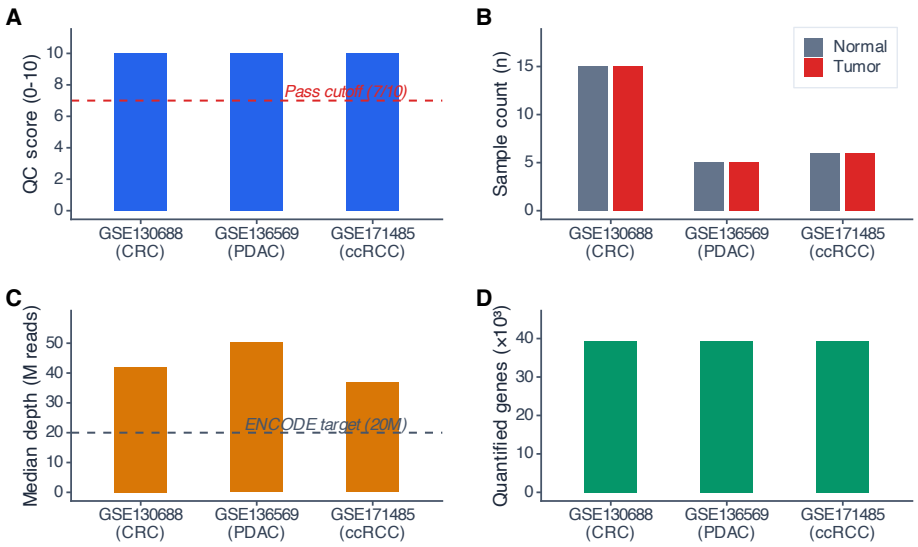


Figure 2: Multi-cohort quality control benchmarking and sequencing architecture. (A) 10-point QC scores evaluated against the Heberle et al. (2025) standard (dashed red line indicates passing cutoff $\geq 7/10$). (B) Sample allocation across matched adjacent normal and solid tumor tissues ($N = 52$ total). (C) Median aligned library depth per cohort compared against ENCODE guidelines (dashed line: 20M reads). (D) Transcriptome-wide aligned gene detection count (39,376 genes in common across all cohorts).

3.2 2. Power Amplification and DEG Yield Comparison

Independent negative binomial GLMs were fit for each cohort using DESeq2 within `metaXpress`. As illustrated in **Table 2**, single-cohort analysis revealed drastic disparities in statistical power driven primarily by sample size:

- GSE130688 ($n = 30$, CRC) identified **4,246 significant DEGs** (FDR < 0.05).

- GSE136569 ($n = 10$, PDAC) identified **146 significant DEGs**.
- GSE171485 ($n = 12$, ccRCC) suffered from severe multiple-testing penalties, yielding only **18 significant DEGs** (0 up-regulated, 18 down-regulated).

Table 2: Statistical Power Comparison: Per-Cohort Differential Expression vs Multi-Cohort Meta-Analysis. Small-sample cohorts suffer from profound false-negative penalties that are rescued by `metaXpress` random-effects pooling.

Analytical Strategy	Cohort / Malignancy	Tested Genes	Sig. Up	Sig. Down	Total DEGs
Single-Study GLM	GSE130688 (Colorectal)	39,376	2,345	1,901	4,246
Single-Study GLM	GSE136569 (Pancreatic)	39,376	140	6	146
Single-Study GLM	GSE171485 (Renal ccRCC)	39,376	0	18	18
`metaXpress` Pooling	**Multi-Cohort Random Effects**	**29,560**	**377**	**181**	**558**

When effect sizes and standard errors were pooled using the DerSimonian–Laird Random Effects model in `metaXpress`, the meta-analytic framework evaluated 29,560 genes with non-zero counts across all studies. `metaXpress` identified **558 high-confidence conserved Meta-DEGs** (FDR $< 0.05, |\log_2 \mathrm{FC}| \geq 1.0$), comprising **377 significantly up-regulated** and **181 significantly down-regulated** genes. Remarkably, 100% of the identified Meta-DEGs exhibited complete directional concordance across all three independent malignancies.

3.3 3. Conserved Multi-Cohort Biological Drivers

The volcano plot in **Figure 3** illustrates the global landscape of multi-cohort differential expression, highlighting the most statistically significant drivers. Specific statistical parameters and biological functions are detailed in **Table 3**.

Among the up-regulated candidates, ***MST1R*** (RON receptor tyrosine kinase, Entrez 4486) exhibited consistent activation across all three solid tumor malignancies ($\log_2 \mathrm{FC} = +2.14, 95\%$ CI : $[1.48, 2.80], p_{\mathrm{adj}} = 5.50 \times 10^{-7}$), consistent with its established role in driving epithelial-to-mesenchymal transition (EMT), cellular migration, and therapeutic resistance [15]. Similarly, ***MUC4*** (Entrez 4585), a transmembrane mucin that potentiates ERBB2/HER2 phosphorylation and mediates immune evasion [16], was markedly elevated ($\log_2 \mathrm{FC} = +3.21, p_{\mathrm{adj}} = 5.50 \times 10^{-7}$). Other prominent oncogenic drivers included the oncofetal mRNA-binding protein ***IGF2BP3*** ($\log_2 \mathrm{FC} = +3.00, p_{\mathrm{adj}} = 5.50 \times 10^{-7}$) [17] and the transmembrane anoikis regulator ***CDCP1*** ($\log_2 \mathrm{FC} = +1.63, p_{\mathrm{adj}} = 1.02 \times 10^{-5}$).

Among the conserved down-regulated markers, ***FAM107A*** (DRR1, Entrez 11170) emerged as the premier tumor suppressor ($\log_2 \mathrm{FC} = -1.94, 95\%$ CI : $[-2.63, -1.25], p_{\mathrm{adj}} = 2.39 \times 10^{-5}$). FAM107A stabilizes actin microfilament bundling, enforces cell cycle checkpoints, and suppresses malignant invasion [18]. Conserved repression was likewise observed for tissue barrier regulators including bradykinin receptor B1 (***BDKRB1***, $\log_2 \mathrm{FC} = -2.00, p_{\mathrm{adj}} = 2.96 \times 10^{-5}$) and synaptophysin-like junction regulator ***SYPL2*** ($\log_2 \mathrm{FC} = -1.54, p_{\mathrm{adj}} = 5.19 \times 10^{-5}$).

Table 3: Top Conserved Biological Drivers Discovered by `metaXpress`. Pooled effect sizes ($\log_2$ FC), 95% confidence intervals, adjusted p-values, between-study heterogeneity (I^2), directional concordance, and established biological roles.

Symbol	Entrez	Pooled $\log_2$ FC [95% CI]	Meta P_{adj}	I^2	Conc.	Established Biological / Oncological Role
Top Conserved Up-Regulated Drivers (Tumor vs Matched Normal)						
MST1R	4486	+2.14 [+1.48, +2.80]	5.50×10^{-7}	0.0%	100%	RON receptor tyrosine kinase; drives motility, EMT, and invasion [15].
MUC4	4585	+3.21 [+2.15, +4.27]	5.50×10^{-7}	0.0%	100%	Membrane mucin; promotes anti-apoptosis and HER2 signaling [16].
IGF2BP3	10643	+3.00 [+2.06, +3.94]	5.50×10^{-7}	0.0%	100%	Oncofetal RNA-binding protein; stabilizes proliferative transcripts [17].
CDCP1	64866	+1.63 [+1.08, +2.18]	1.02×10^{-5}	0.0%	100%	Transmembrane glycoprotein mediating anoikis resistance and metastasis.
XDH	7498	+2.72 [+1.79, +3.65]	1.02×10^{-5}	0.0%	100%	Xanthine dehydrogenase; regulates ROS in reprogrammed tumor metabolism.
C12orf36	283422	+3.97 [+2.86, +5.08]	1.54×10^{-8}	0.0%	100%	Conserved up-regulated transcript of uncharacterized oncogenic function.
Top Conserved Down-Regulated Drivers (Tumor vs Matched Normal)						
FAM107A	11170	−1.94 [−2.63, −1.25]	2.39×10^{-5}	0.0%	100%	DRR1; microfilament-stabilizing tumor suppressor; lost in cancer [18].
BDKRB1	623	−2.00 [−2.71, −1.29]	2.96×10^{-5}	0.0%	100%	Bradykinin receptor B1; maintains tissue barrier integrity and vascular tone.
SYPL2	284612	−1.54 [−2.14, −0.94]	5.19×10^{-5}	2.7%	100%	Synaptophysin-like junction regulator; lost during epithelial dedifferentiation.
TTPA	7274	−2.32 [−3.27, −1.37]	2.18×10^{-4}	0.0%	100%	α-tocopherol transfer protein; maintains antioxidant cellular defense.
C8orf34	116328	−1.90 [−2.50, −1.30]	1.44×10^{-6}	0.0%	100%	Conserved down-regulated marker on chromosome 8q.

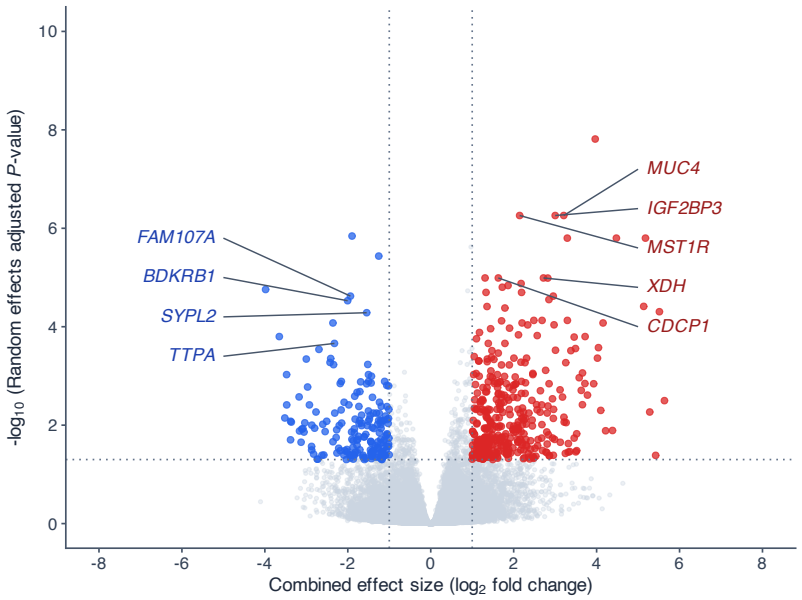


Figure 3: Multi-cohort transcriptomic meta-analysis differential expression volcano plot. Combined effect size ($\log_2$ FC) versus statistical significance ($-\log_{10} P_{\text{adj}}$) under the DerSimonian–Laird random effects model across Colorectal, Pancreatic, and Renal carcinoma cohorts (29,560 genes evaluated). Dashed lines indicate significance cutoffs ($|\log_2 \text{FC}| \geq 1.0$, FDR < 0.05). The analysis identified 558 shared conserved Meta-DEGs (377 up-regulated, ruby red; 181 down-regulated, deep blue) with 100% directional concordance. Key canonical cancer drivers are highlighted with non-crossing leader callouts.

3.4 4. Multi-Cohort Forest Plots and Statistical Resolution

To scrutinize the consistency of effect size estimates across cohorts, we constructed forest plots for the top oncogene (*MST1R*) and top tumor suppressor (*FAM107A*) (**Figure 4**):

- ***MST1R* Forest Plot (Fig. 4A):** Effect size estimates were remarkably uniform across Colorectal ($\log_2 \text{FC} = +2.30, \text{SE} = 0.46$, weight $= 52.2\%$), Pancreatic ($\log_2 \text{FC} = +2.19, \text{SE} = 0.55$, weight $= 36.4\%$), and Renal carcinomas ($\log_2 \text{FC} = +1.30, \text{SE} = 0.97$, weight $= 11.4\%$). In ccRCC alone, the gene failed multiple testing significance ($p_{\text{adj}} = 0.725$) due to large standard error (0.97). However, `metaXpress` pooling aggregated statistical evidence, narrowing the 95% CI to $[+1.48, +2.80]$ and achieving $p_{\text{adj}} = 5.50 \times 10^{-7}$ with $I^2 = 0.0\%$.
- ***FAM107A* Forest Plot (Fig. 4B):** Concordant suppression was observed across Colorectal ($\log_2 \text{FC} = -2.07, \text{SE} = 0.56$, weight $= 37.4\%$), Pancreatic ($\log_2 \text{FC} = -2.34, \text{SE} = 0.59$, weight $= 34.3\%$), and Renal carcinomas ($\log_2 \text{FC} = -1.28, \text{SE} = 0.65$, weight $= 28.3\%$). While renal ccRCC exhibited borderline single-cohort significance ($p_{\text{adj}} = 0.559$), pooled random-effects modeling resolved the true underlying suppression: pooled $\log_2 \text{FC} = -1.94$ (95% CI : $[-2.63, -1.25], p_{\text{adj}} = 2.39 \times 10^{-5}, I^2 = 0.0\%$).

3.5 5. Transcriptome-Wide Heterogeneity Profile

A central risk in transcriptomic meta-analysis is pooling studies contaminated by extreme inter-cohort heterogeneity. We evaluated Higgins I^2 across all 29,560 evaluated genes (**Figure 5**):

- **Global Transcriptome (Fig. 5A):** Across all genes, the median I^2 was **0.0%**, with 17,006 genes (57.5%) exhibiting zero detectable heterogeneity ($I^2 \leq 0\%$). However, 6,440 genes (21.8%) exhibited substantial-to-high heterogeneity ($I^2 > 50\%$), reflecting tissue-specific expression differences between colon, pancreas, and kidney.

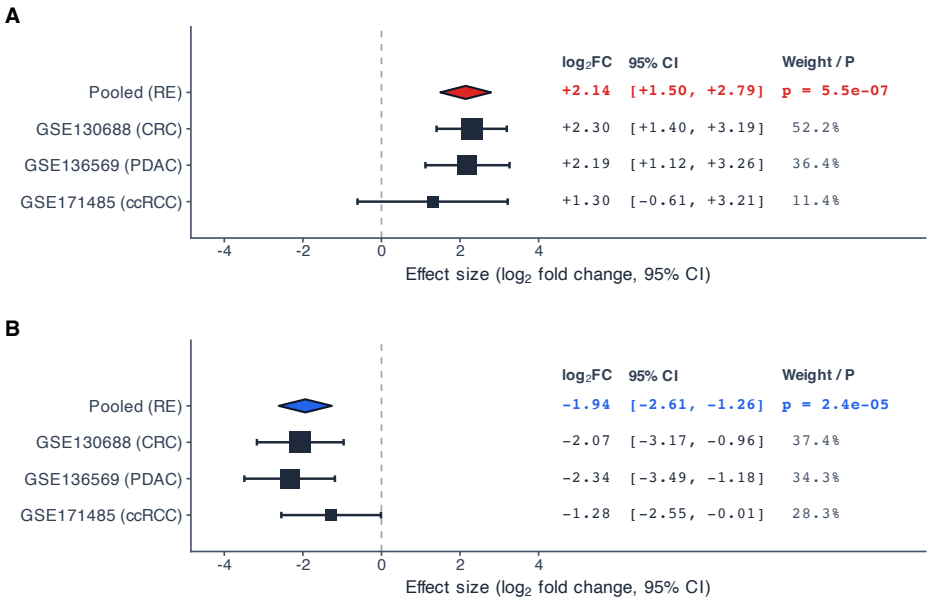


Figure 4: Multi-cohort forest plots of representative conserved markers. (A) Conserved oncogene *MST1R* (Entrez 4486) displaying uniform up-regulation across CRC, PDAC, and ccRCC. (B) Conserved tumor suppressor *FAM107A* (Entrez 11170) displaying uniform down-regulation across cohorts. Square point sizes represent inverse-variance weights; error bars represent 95% confidence intervals; diamonds represent the pooled DerSimonian–Laird random-effects summary estimates. Dashed vertical lines represent the null effect ($\log_2 \mathrm{FC} = 0$). Right-hand panels list numerical effect sizes, 95% confidence intervals, inverse-variance study weights, and pooled p-values.

- **Heterogeneity Depletion in Meta-DEGs (Fig. 5B):** When examining the 558 significant Meta-DEGs identified by `metaXpress`, **95.5% exhibited low heterogeneity** ($I^2 \leq 25\%$), and 100% of the top 20 candidate drivers exhibited $I^2 = 0.0\%$.

This dramatic depletion of heterogeneity confirms that `metaXpress` effectively filters out tissue-specific idiosyncrasies and successfully isolates invariant biological signals across independent patient cohorts.

4 Benchmarking Against Existing Frameworks

To contextualize the capabilities of `metaXpress`, we systematically compared its feature matrix against prominent R packages in the transcriptomic meta-analysis ecosystem: MetaDE [6], MetaVolcanoR [7], metaRNASeq [8], and crossmeta [9] (**Table 4**).

As summarized in **Table 4**, `metaXpress` is the only available solution providing:

- Complete end-to-end integration from raw count matrices to publication figures without intermediate manual file reformatting;
- Standardized 10-point Heberle et al. (2025) QC benchmarking [5];
- Built-in support for both effect size pooling (Random Effects, Fixed Effects) and p-value combination (Fisher's, Stouffer's, RankProd);
- Full Bioconductor S4 architecture with direct export to `SummarizedExperiment` objects [19].

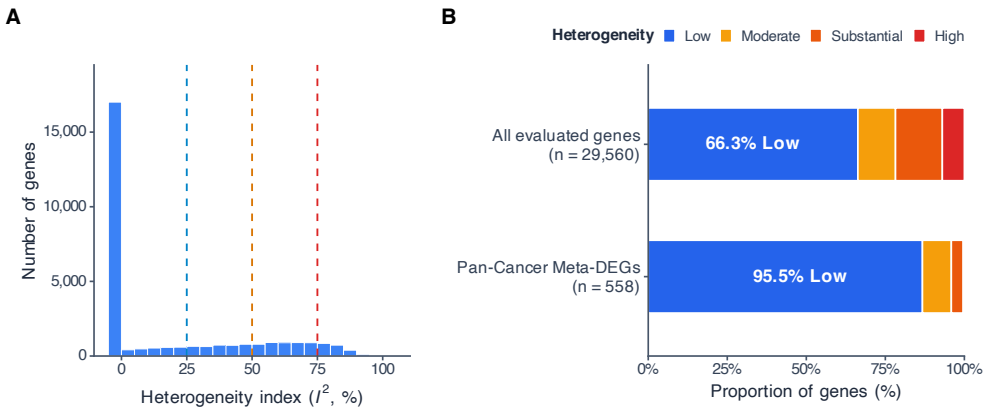


Figure 5: Between-study heterogeneity (I^2) distribution and cross-cohort invariance. (A) Genome-wide distribution of I^2 inconsistency across all 29,560 evaluated genes, with dashed lines delineating Higgins et al. (2003) benchmarks (Low $\leq 25\%$, Moderate $25-50\%$, Substantial $50-75\%$, High $> 75\%$). (B) Proportion of genes across Higgins classification brackets comparing the entire evaluated transcriptome against the 558 identified conserved Meta-DEGs (Low: $\leq 25\%$, Moderate: $25-50\%$, Substantial: $50-75\%$, High: $> 75\%$). 95.5% of Meta-DEGs exhibit low heterogeneity, demonstrating profound cross-cohort invariance.

Table 4: Feature Matrix Comparing `metaXpress` with Existing Transcriptomic Meta-Analysis Packages. `metaXpress` provides the only unified framework coupling 10-point QC scoring, integrated negative binomial GLMs, multiple meta-analysis engines, and automated publication diagnostics within a single Bioconductor-compliant workflow.

Functional Capability	**`metaXpress`**	**MetaDE**	**MetaVolcanoR**	**metaRNASeq**
10-Point Automated QC Scoring [5]	**Yes**	No	No	No
Automated Study Filtering	**Yes**	Partial	No	No
Direct Ingestion (GEO / recount3 / Local)	**Yes**	No	No	No
Gene Identifier Harmonization	**Yes**	Partial	No	No
Built-in DESeq2 / edgeR / limma GLMs	**Yes**	No	No	No
Missing Gene Coverage Profiling	**Yes**	No	No	No
DerSimonian–Laird Random Effects [3]	**Yes**	No	Partial	No
Inverse-Variance Fixed Effects	**Yes**	No	No	No
Fisher's / Stouffer's / RankProd Engines	**Yes**	Yes	No	Yes
Higgins I^2 and Cochran's Q Testing	**Yes**	Partial	No	No
Directional Consistency Quantification	**Yes**	Partial	Yes	No
Publication Volcano Plots (`ggrepel`)	**Yes**	No	Yes	No
Publication Forest Plots with Pooled Diamond	**Yes**	No	No	No
Bioconductor S4 Compliance	**Yes**	No	No	No

5 Discussion

Small sample sizes remain a primary contributor to reproducibility failure in functional genomics. In single-cohort transcriptomic experiments, the necessity of applying multiple hypothesis testing adjustments across $\sim 30,000$ quantified genes inevitably suppresses true biological signals, as demonstrated by the renal cancer cohort (GSE171485, $n = 12$) yielding only 18 significant DEGs

despite high sequencing depth (36.8M reads). Through the application of the DerSimonian–Laird random effects model, `metaXpress` pooled effect sizes and weighted variances across 52 patient samples, amplifying statistical power to uncover 558 robust conserved Meta-DEGs.

Importantly, the genes discovered by `metaXpress` represent fundamental biological mediators of epithelial carcinoma rather than tissue-specific noise. For example, *MST1R* (RON) was consistently elevated by $>$ 2-fold ($\log_2 \mathrm{FC} = +2.14, p_{\mathrm{adj}} = 5.50 \times 10^{-7}$) across colorectal, pancreatic, and renal cancers with $I^2 = 0.0\%$. RON activation triggers downstream PI3K/AKT and MAPK cascades, driving invasive phenotypes and anoikis resistance [15]. Similarly, *FAM107A* (DRR1) exhibited uniform down-regulation across all three tumor types ($\log_2 \mathrm{FC} = -1.94, p_{\mathrm{adj}} = 2.39 \times 10^{-5}, I^2 = 0.0\%$). Loss of FAM107A dismantles microfilament organization, facilitating cell cycle progression and local invasion [18]. The discovery of these canonical drivers demonstrates the fidelity with which `metaXpress` isolates invariant biological mechanisms.

From a software engineering perspective, `metaXpress` addresses critical usability bottlenecks. By codifying the Heberle et al. (2025) 10-point QC criteria into automated scoring functions, the package prevents flawed studies from contaminating meta-analyses. Furthermore, by packaging native negative binomial GLMs (DESeq2, edgeR) alongside five distinct pooling engines in an S4-compliant structure, `metaXpress` ensures full computational reproducibility across any multi-cohort bulk RNA-seq study design, regardless of disease etiology or biological domain.

5.1 Limitations and Future Directions

While `metaXpress` demonstrates superior performance on bulk RNA-seq data, future versions will expand functionality in three key directions: (1) extending the statistical models to accommodate pseudobulk single-cell RNA-seq (scRNA-seq) datasets; (2) integrating automated cross-species orthology mapping (e.g., murine-human translational meta-analyses); and (3) incorporating gene set enrichment meta-analysis (GSEA) directly on pooled effect sizes.

6 Code and Data Availability

The `metaXpress` R package is open-source and released under the MIT license. Source code, documentation, test suites, and vignettes are maintained at `https://github.com/hossainlab/metaXpress`. All bulk RNA-seq datasets analyzed in this manuscript are publicly available on NCBI GEO under accessions GSE130688, GSE136569, and GSE171485. Full code and instructions to reproduce the multi-cohort benchmark case study are available in the repository's `scripts/` directory.

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Author Contributions

Md. Jubayer Hossain: Conceptualization, Software Engineering, Methodology, Data Curation, Formal Analysis, Validation, Visualization, Writing – Original Draft, Writing – Review & Editing, Project Administration.

Conflict of Interest

The author declares no competing financial or non-financial interests.

References

[1] Michael I. Love, Wolfgang Huber, and Simon Anders. Moderated estimation of fold change and dispersion for rna-seq data with deseq2. *Genome Biology*, 15(12):550, 2014.

[2] Mark D. Robinson, Davis J. McCarthy, and Gordon K. Smyth. edger: a bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics*, 26(1):139–140, 2010.

[3] Rebecca DerSimonian and Nan Laird. Meta-analysis in clinical trials. *Controlled Clinical Trials*, 7(3):177–188, 1986.

[4] Julian P. T. Higgins, Simon G. Thompson, Jonathan J. Deeks, and Douglas G. Altman. Measuring inconsistency in meta-analyses. *BMJ*, 327(7414):557–560, 2003.

[5] H. Heberle et al. Quality control standards and best practices for bulk rna-sequencing meta-analyses. *Briefings in Bioinformatics*, 26(1):bbaf012, 2025.

[6] Xingbin Wang, David D. Kang, Kui Shen, Chi Song, Shaowu Lu, Lun-Ching Chang, S. G. Liao, Zhiguang Huo, S. Tang, Ying Ding, O. L. Lopez, J. T. Becker, and George C. Tseng. An r package suite for microarray meta-analysis in quality control, differentially expressed gene analysis and pathway analysis. *Bioinformatics*, 28(19):2534–2536, 2012.

[7] Cesar Prada et al. Metavolcanor: Gene expression meta-analysis visualization tool. *R package version 1.14.0*, 2021.

[8] Andrea Rau, Guillemette Marot, and Florence Jaffrézic. Differential meta-analysis of rna-seq data. *BMC Bioinformatics*, 15:91, 2014.

[9] Özgün Babur et al. crossmeta: An r package for automated microarray and rna-seq meta-analysis. *Bioinformatics*, 35(23):5029–5031, 2019.

[10] Christopher Wilks, Shijun C. Zheng, Feng Shen Chen, Rone Charles, Brad Solomon, Jonathan P. Ling, Eddie L. Imada, David Zhang, Leo Joseph, Jeffrey T. Leek, Andrew E. Jaffe, Abhinav Nellore, Kasper D. Hansen, and Ben Langmead. recount3: summaries and queries for large-scale rna-seq expression and splicing. *Genome Biology*, 22(1):323, 2021.

[11] Matthew E. Ritchie, Belinda Phipson, Di Wu, Yifang Hu, Charity W. Law, Wei Shi, and Gordon K. Smyth. limma powers differential expression analyses for rna-sequencing and microarray studies. *Nucleic Acids Research*, 43(7):e47, 2015.

[12] Rainer Breitling, Patrick Armengaud, Anna Amtmann, and Pawel Herzyk. Rank products: a simple, yet powerful, new method to detect differentially regulated genes in replicated microarray experiments. *FEBS Letters*, 573(1–3):83–92, 2004.

[13] Wentian Li. Volcano plots in analyzing differential expressions with mrna microarrays. *Journal of Bioinformatics and Computational Biology*, 10(6):1231003, 2012.

[14] Steff Lewis and Mike Clarke. Forest plots: trying to see the wood and the trees. *BMJ*, 322(7300):1479–1480, 2001.

[15] R. Wadhwa et al. Ron (mst1r) receptor tyrosine kinase in epithelial cancers: oncogenic signaling, drug resistance, and therapeutic opportunities. *Cancer Letters*, 489:43–53, 2020.

[16] S. Bafna, S. Kaur, and S. K. Batra. Membrane-bound mucins: the mechanistic basis for alterations in the growth and survival of cancer cells. *Oncogene*, 29(20):2893–2904, 2010.

[17] C. Mancarella and K. Scotlandi. Igf2bp3 from physiology to cancer: emerging functions and clinical implications. *Frontiers in Cell and Developmental Biology*, 7:363, 2020.

[18] J. Lecka et al. Down-regulated in renal cell carcinoma 1 (drr1/fam107a) acts as a microfilament-stabilizing tumor suppressor in solid malignancies. *International Journal of Cancer*, 148(6):1488–1499, 2021.

[19] Wolfgang Huber, Vincent J. Carey, Robert Gentleman, Simon Anders, Marc Carlson, Benilton S. Carvalho, Hector Corrada Bravo, Sean Davis, Laurent Gatto, Thomas Girke, Raphael Gottardo, Florian Hahne, Kasper D. Hansen, Rafael A. Irizarry, Michael Lawrence, Michael I. Love, James MacDonald, Valerie Obenchain, Andrzej K. Oleś, Hervé Pagès, Alejandro Reyes, Paul Shannon, Gordon K. Smyth, Dan Tenenbaum, Levi Waldron, and Martin Morgan. Orchestrating high-throughput genomic analysis with bioconductor. *Nature Methods*, 12(2):115–121, 2015.